15 Implement Iris Flower Classification using Naive Bayes classifier
Code:

```
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

# Load dataset
iris = load_iris()
X = iris.data
y = iris.target

# Standardize the data
X_scaled = StandardScaler().fit_transform(X)

# Reduce 4 features to 2 principal components
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

print("Principal Component Analysis (PCA)")
print("Original Number of Features:", X.shape[1])
print("Reduced Number of Features:", X_pca.shape[1])
print("Variance Explained:",
      round(sum(pca.explained_variance_ratio_) * 100, 2), "%")

# Plot graph
for class_id, name in enumerate(iris.target_names):
    plt.scatter(
        X_pca[y == class_id, 0],
        X_pca[y == class_id, 1],
        label=name
    )

plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("PCA - Iris Dataset")
plt.legend()
plt.grid(True)
plt.show()
```

Output:

```
Principal Component Analysis (PCA)
Original Number of Features: 4
Reduced Number of Features: 2
Variance Explained: 95.81 %
```

